

A close-up, top-down view of a large pile of brown wooden yo-yos. Each yo-yo has a blue ribbon wrapped around its central axle and an orange string tied to it. The yo-yos are stacked and scattered, creating a textured, repetitive pattern of brown wood, blue plastic, and orange cord. The lighting is warm, highlighting the grain of the wood and the sheen of the plastic.

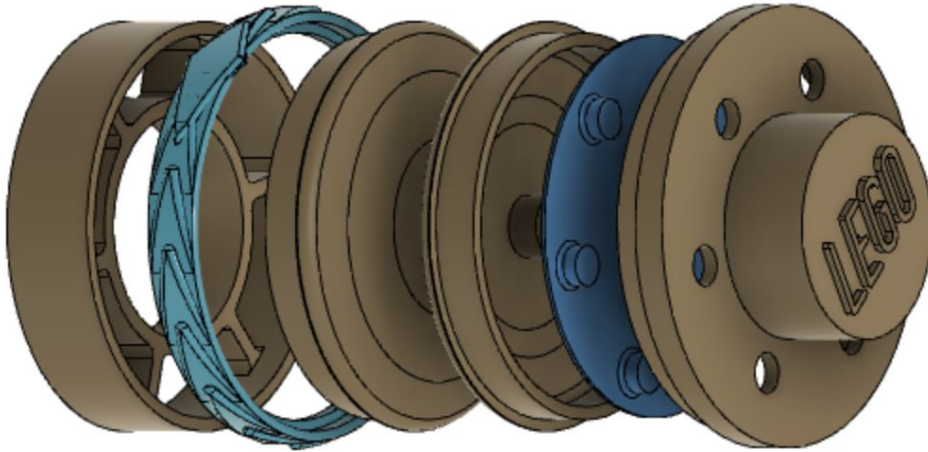
YoNinjagYo

Adi, Hannah, Kiet, Laura, Rethabile



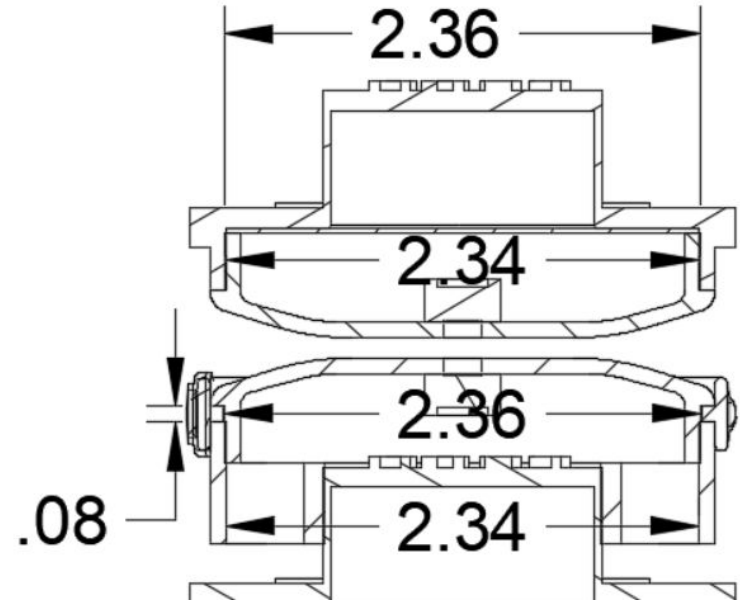
Thanks Dr. Ramos!

Design Specifications



Must stay together when dropped!

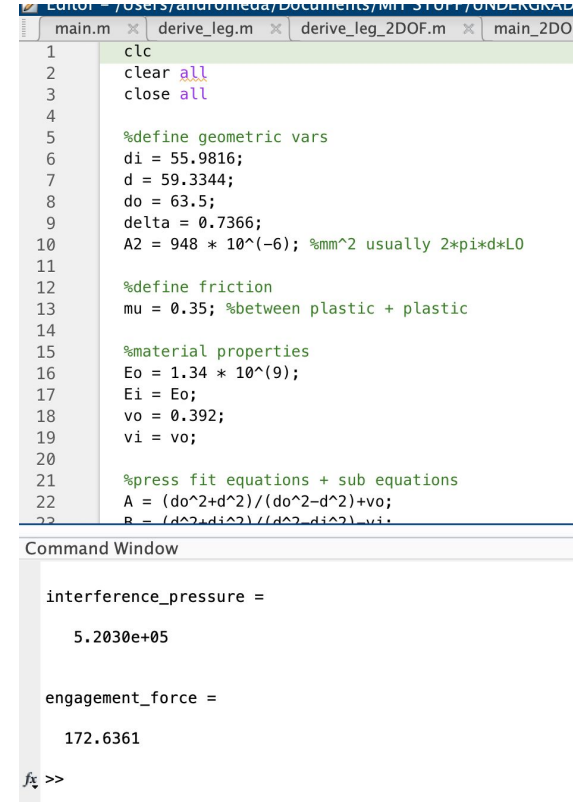
To ensure good press fit:



Tolerance: 0.001

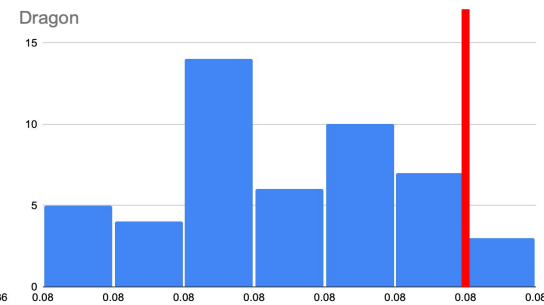
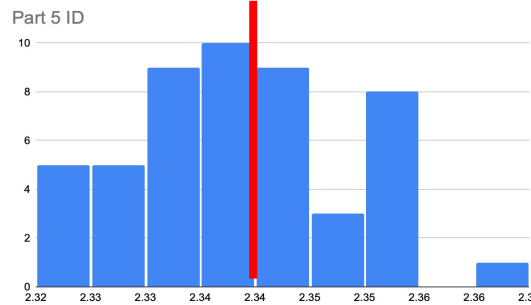
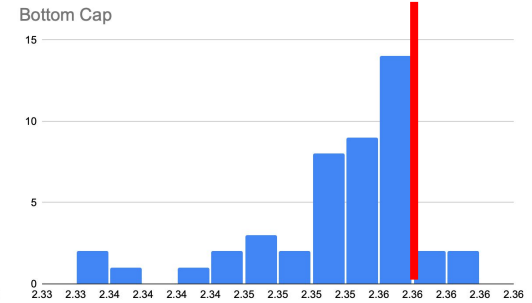
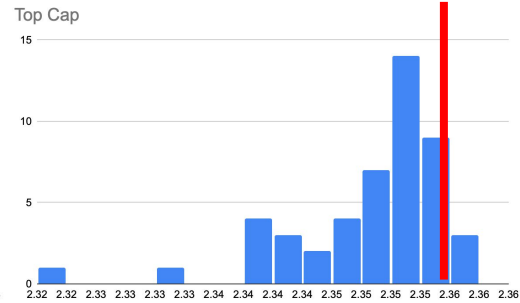
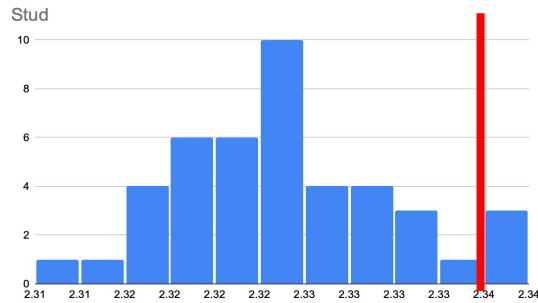
Design Specifications - Press Fit Calculations & Testing

- Used Interference from TA Poster
- Put press fit calcs
 - Tried to maintain a length of engagement of 0.2 between all press fitted parts
 - Included making the back part longer
- Initially we wanted variation of .001 tops
- We later learned this math wasn't accurate



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Editor - 70users/andromeda/Documents/MAT STUFF/UNDERGRAD
main.m x derive_leg.m x derive_leg_2DOF.m x main_2DO
1 clc
2 clear all
3 close all
4
5 %define geometric vars
6 di = 55.9816;
7 d = 59.3344;
8 do = 63.5;
9 delta = 0.7366;
10 A2 = 948 * 10^(-6); %mm^2 usually 2*pi*d*L0
11
12 %define friction
13 mu = 0.35; %between plastic + plastic
14
15 %material properties
16 Eo = 1.34 * 10^(9);
17 Ei = Eo;
18 vo = 0.392;
19 vi = vo;
20
21 %press fit equations + sub equations
22 A = (do^2+d^2)/(do^2-d^2)+vo;
23 B = (do^2+di^2)/(do^2-di^2)-vi;
24
25 interference_pressure =
26     5.2030e+05
27
28 engagement_force =
29     172.6361
30
31 fx >>
```

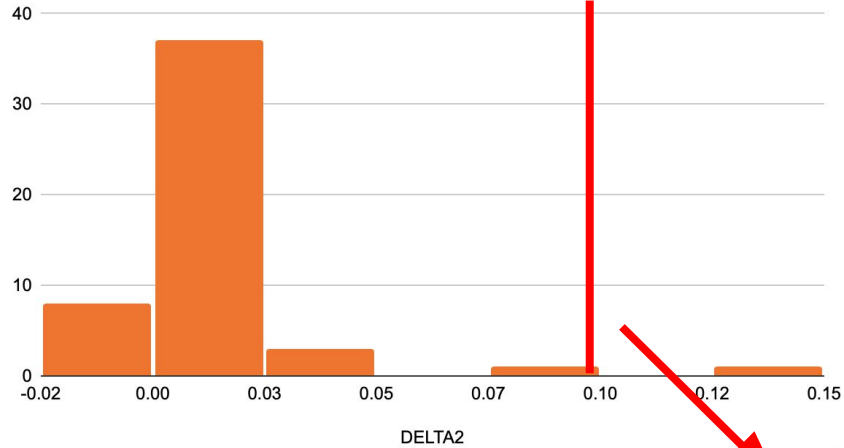
Variations... Not $\pm .001$ (at least 7 days after assembly)



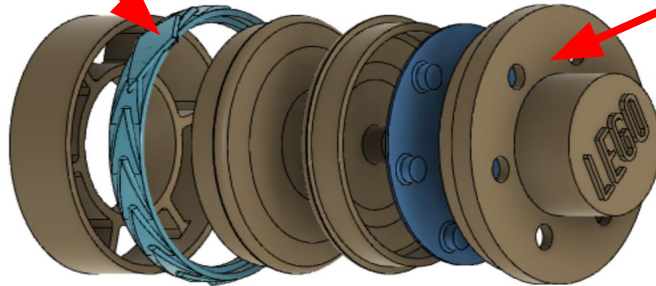
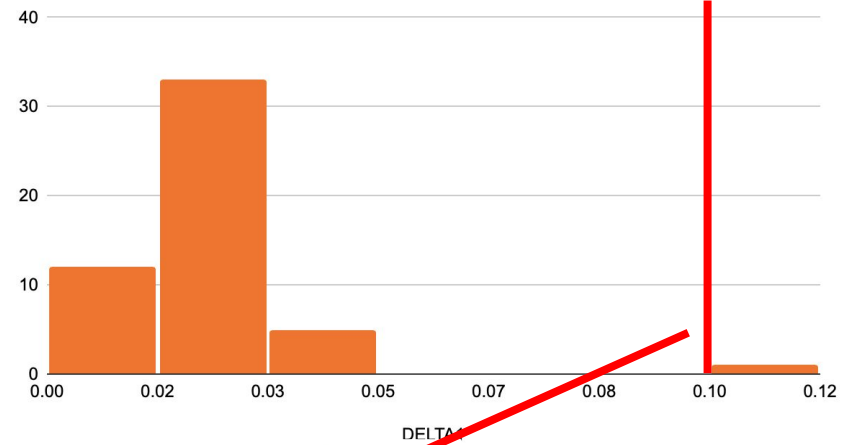
	Stud	Top Cap	Bottom Cap	Part 5	Dragon
Design	2.34	2.36	2.36	2.34	0.08
Actual	2.3237 $\pm$ 0.0041	2.3484 $\pm$ 0.0020	2.3554 $\pm$ 0.0064	2.3382 $\pm$ 0.0028	0.07915 $\pm$ 0.00051

Variations in Interference

Engagement between "Back" and Yoyo "Body"



Engagement between "Stud" and Yoyo "Body"



Process Optimization - Avoiding Defects

1	Trial	Heating Time	Forming Time	Fan delay	Acceptable?	Notes
2	1	11	15	15	Y	
3	2	11	10	5	Y	
4	3	11	5	5	N	too soft becomes dome shaped --> try decreasing temp (8s?)
5	4	8	5	0	N	too soft needs more cooling time --> NO POINTY
6	5	8	8	0	N	needs more cooling time --> NO POINTY
7	6	10	10	2	N	little domy but point OK
8	7	11	12	2	Y	
9	8	10	8	1	N	--> LOL we turned the fans on --> no fan delay? still tiny bit domy
10	9	10	9	0	N	--> still more cooling time
11	10	10	11	0	Y	--> we're happy

1	Trial	Injection Pressure (psi)	Dosage Stroke (in)	Holding Pressure (psi)	Cooling Time (s)	Temperature (F)	Ejector pins length	Result
2	1	2800	3.2	All 4 at 800	15	430(all4)	5.375	didn't get pushed out all the way and also experienced flash
3	2		2.5		25			still flash (more than before), and the runner detached
4	3	2000	3.2		15			less flash, runner detached (still in mold)
5	4-1	1500						more flash than before! we can't control how temperature of the mold is increasing
6	4-2							pretty similar to above
7	5	1000						least amount of flash so far!
8	6	500						no flash! First time using pliers to remove
9	7				20			no flash, but needed pliers
10	8				30			"
11	9				20		5.5	no change
12	10				30			short shot! but it came out a little more it seems the ejector pins might be blocking the flow?
13	11						5.375 trying mold release only on one side	stuck in the cavity lol
14	12						mold release on both sides	HE CAME OUT!!!
15	13 and 14							short shot (strange)
16	15	750			15			came out, but tiny bit of flash, bent
17	16	600			20			slightest bit of flash on runner, but c
18	17						added mold release	didn't come out (mold too hot?)
19								



Process Optimization - Assembly Plan



- Heat gun is used to assemble the press fit
- Mix-and-match parts to form compatible press fit pair



Major Lessons

- Rate:
 - Having thinner parts parts helped greatly with the rate of production
 - More parts (complexity) increase both design and production (manuf. + asm.) time
- Quality:
 - The enemy of rate, spending more time during machining could give parts with better finish
 - Creep was also the enemy of the quality of our press fits – look out for such confounding factors in design
- Cost:
 - The yo-yos were free! For us. But if we were to scale up, reducing the material used in the design and dropping our rates would have been really beneficial.
- Flexibility:
 - Even simple changes can take significant amount of time to implement

Thank You!

